DSA06 - Data Handling and Visualization
Activity 7: Correlation Detective

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Course Code / Course Name	DSA0602 / Data Handling and Visualization
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Scenario: You are the data detective hired by the product team of a mobile app. The file holds 14 daily records from the beta month, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Case File CD-D06: The Mobile app lab case (Tool: R, ggplot2 / base R).

Evidence: the case file was created by running the code below in R without any change (set.seed(543939)).

```
set.seed(543939)
n <- 14
evening_index <- round(rnorm(n, 47.6, 13.6), 1)
notifications <- round(38.1 + 9.7 * (0.93 * (evening_index - 47.6) / 13.6 + 0.38 * rnorm(n)), 1)
sessions <- round(23.8 + 5.8 * (0.86 * (evening_index - 47.6) / 13.6 + 0.50 * rnorm(n)), 1)
bright_dev <- round(seq(-46.3, 46.3, length.out = n)[order(runif(n))], 1)
eye_strain <- round(3.8 + 11.1 * (bright_dev / 46.3)^2 + rnorm(n, 0, 0.4), 0)
testers <- round(rnorm(n, 114, 24.4), 0)
crash_reports <- round(14 + 4.9 * (0.10 * (testers - 114) / 24.4 + 0.99 * rnorm(n)), 0)
testers[8] <- 195; crash_reports[8] <- 29   # row 8 was typed in by a trainee
case <- data.frame(evening_index, notifications, sessions, bright_dev,
                   eye_strain, testers, crash_reports)
```

Case file data (output of print(case)):

#	evening_index	notifications	sessions	bright_dev	eye_strain	testers	crash_reports
1	67.1	49.9	33.9	-32.1	9	74	16
2	70.7	59.3	28.3	-39.2	12	126	19
3	57.7	49.8	28.4	-10.7	4	47	15
4	73.2	49.5	32.9	46.3	15	107	18
5	64.9	58	33.8	-24.9	7	124	16
6	46.4	40.5	26.4	24.9	7	92	15
7	55.3	46.2	29.2	-17.8	6	100	14
8	34.8	27.9	17.2	39.2	12	195	29
9	53.8	46.4	27.6	17.8	5	111	7
10	42.1	39.5	17.8	-46.3	14	137	16
11	14.9	18.5	10.4	-3.6	4	97	9
12	60.1	52.7	27.2	32.1	9	118	16
13	56.4	45.2	24.8	10.7	5	110	20
14	23.1	15.6	15.6	3.6	4	118	15

Detective's rule: correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.

TASK 1: The matrix

Use cor(case) to get the 7 × 7 correlation matrix (rounded to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot).

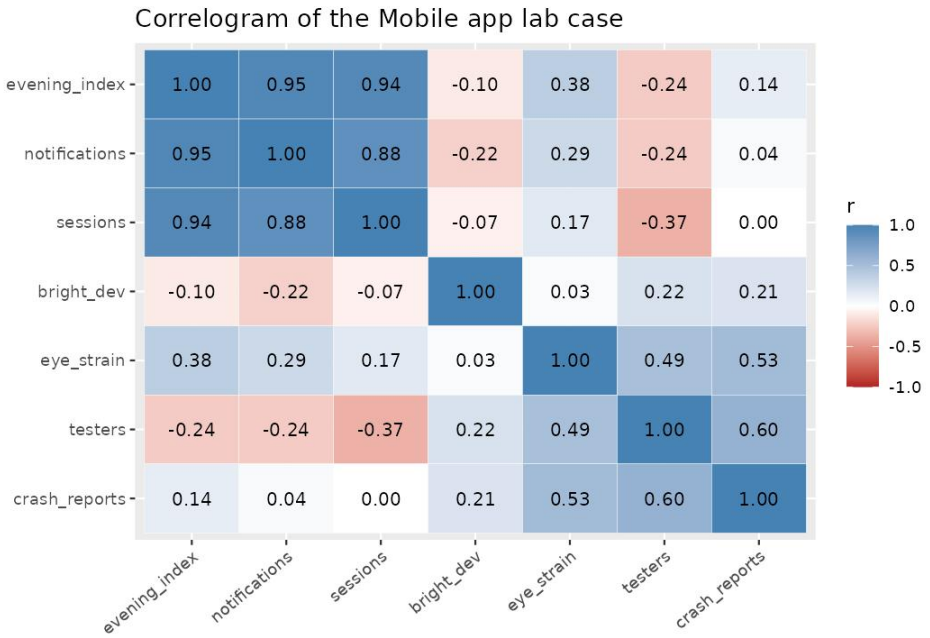
```
library(ggplot2)
cm <- round(cor(case), 2)          # 7 x 7 correlation matrix
print(cm)
nrow(case)                         # n
choose(ncol(case), 2)              # distinct pairs

cd <- as.data.frame(as.table(cm))
names(cd) <- c("Var1", "Var2", "r")
cd$Var2 <- factor(cd$Var2, levels = rev(colnames(cm)))
ggplot(cd, aes(Var1, Var2, fill = r)) +
  geom_tile(color = "white") +
  geom_text(aes(label = sprintf("%.2f", r)), size = 3.4) +
  scale_fill_gradient2(low = "firebrick", mid = "white", high = "steelblue",
                       midpoint = 0, limits = c(-1, 1)) +
  labs(title = "Correlogram of the Mobile app lab case", x = NULL, y = NULL) +
  theme(axis.text.x = element_text(angle = 40, hjust = 1))
```

OUTPUT:

	evening_index	notifications	sessions	bright_dev	eye_strain	testers	crash_reports
evening_index	1.00	0.95	0.94	-0.10	0.38	-0.24	0.14
notifications	0.95	1.00	0.88	-0.22	0.29	-0.24	0.04
sessions	0.94	0.88	1.00	-0.07	0.17	-0.37	0.00
bright_dev	-0.10	-0.22	-0.07	1.00	0.03	0.22	0.21
eye_strain	0.38	0.29	0.17	0.03	1.00	0.49	0.53
testers	-0.24	-0.24	-0.37	0.22	0.49	1.00	0.60
crash_reports	0.14	0.04	0.00	0.21	0.53	0.60	1.00

Output 1a: Correlation matrix of the case file (rounded to 2 decimals)



Output 1b: Correlogram of the Mobile app lab case

Ans:

n = 14 daily records. There are 7 variables, so the number of distinct pairs is 7 × 6 ÷ 2 = 21 (the 7 diagonal values of 1.00 and the mirror-image half of the matrix are not counted). In the correlogram, the dark blue

block at the top left (evening_index, notifications, sessions) stands out at once, while most of the other cells are pale.

TASK 2: The strongest clues

Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7).

```
ut <- which(upper.tri(cm), arr.ind = TRUE)      # off-diagonal pairs only
pairs_df <- data.frame(var1 = rownames(cm)[ut[, 1]],
                       var2 = colnames(cm)[ut[, 2]],
                       r    = cm[ut])
pairs_df[order(-pairs_df$r), ][1, ]                  # strongest positive
pairs_df[order(pairs_df$r), ][1, ]                   # strongest negative
```

OUTPUT:

Clue	Pair of variables	r	Strength
Strongest positive	evening_index and notifications	0.95	Strong (above 0.7)
Strongest negative	sessions and testers	−0.37	Moderate (0.3 to 0.7)

Output 2: Strongest positive and strongest negative pair (off-diagonal)

Ans:

The strongest positive pair is evening_index and notifications (r = 0.95, strong). The strongest negative pair is sessions and testers (r = −0.37, moderate). Two more pairs are also strong: evening_index and sessions (0.94) and notifications and sessions (0.88), which is the first sign that these three variables move together.

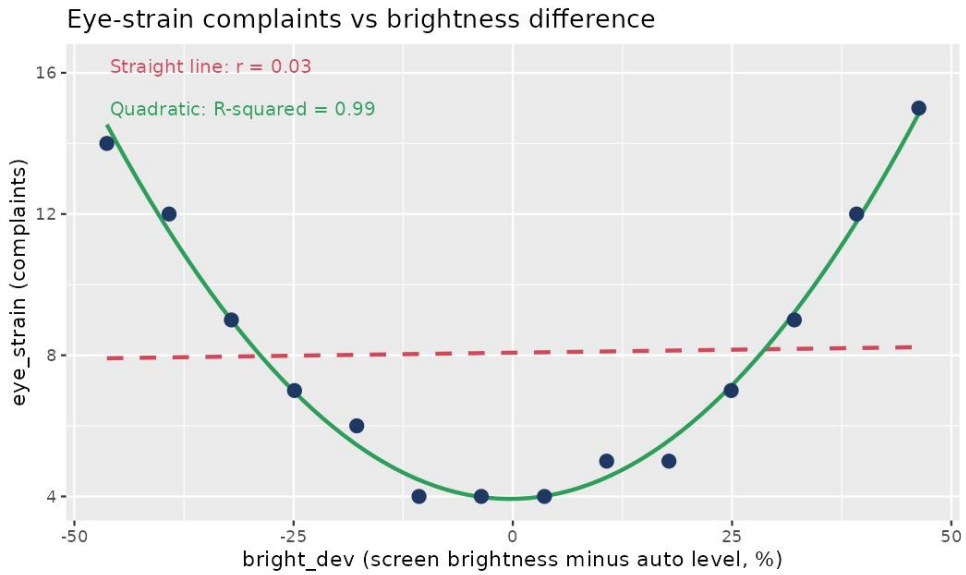
TASK 3: The deceptive zero

Plot eye_strain against bright_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship.

```
cor(case$bright_dev, case$eye_strain)                       # straight-line r
quad <- lm(eye_strain ~ bright_dev + I(bright_dev^2), data = case)
summary(quad)$r.squared                                     # R-squared of the quadratic

ggplot(case, aes(bright_dev, eye_strain)) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE,
              colour = "#D1495B", linetype = "dashed") +
  geom_smooth(method = "lm", formula = y ~ x + I(x^2), se = FALSE,
              colour = "#2E9E5B") +
  geom_point(size = 3, colour = "#1F3864") +
  labs(title = "Eye-strain complaints vs brightness difference",
       x = "bright_dev", y = "eye_strain")
```

OUTPUT:



Output 3: eye_strain vs bright_dev with straight-line and quadratic fits

Measure	Value
r (straight line)	0.03 (essentially zero)
R-squared (quadratic fit)	0.99
Fitted curve	eye_strain = 3.93 + 0.0034 × bright_dev + 0.0050 × bright_dev²

Ans:

r = 0.03, but the quadratic fit has R-squared = 0.99, so the link is almost perfect. r misses it because the relationship is U-shaped: eye strain rises when brightness moves far from the auto level in either direction, so the falling half and the rising half cancel each other and r, which measures only a straight-line link, comes out near zero.

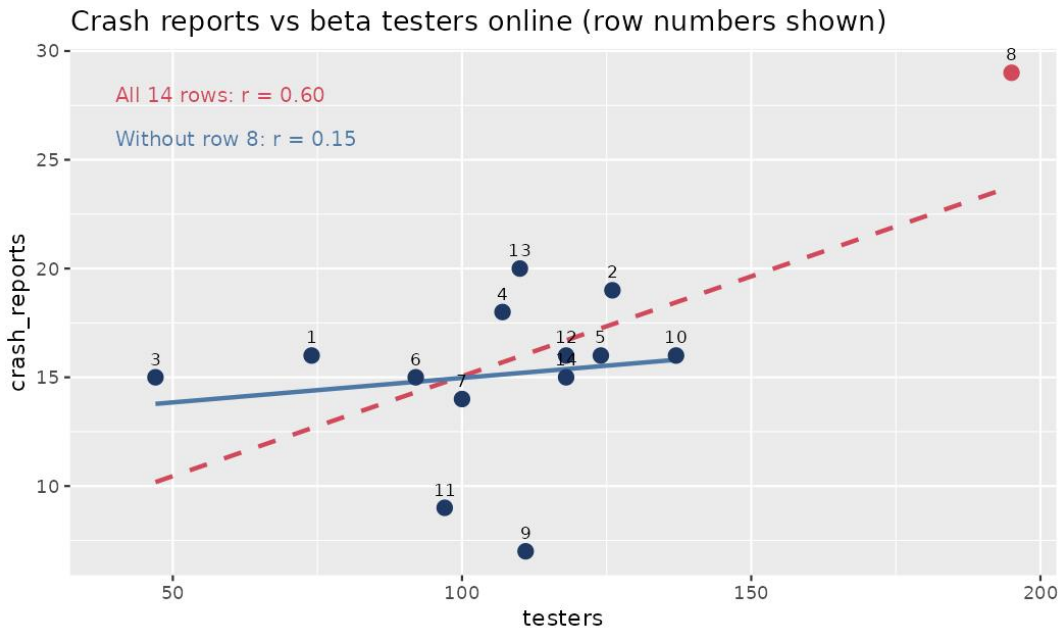
TASK 4: The outlier alibi

Plot crash_reports against testers. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r.

```
case$row <- seq_len(nrow(case))
cor(case$testers, case$crash_reports)               # all 14 rows
cor(case$testers[-8], case$crash_reports[-8])       # without row 8
case[8, ]                                           # the suspicious record

ggplot(case, aes(testers, crash_reports)) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE,
              colour = "#D1495B", linetype = "dashed") +
  geom_smooth(data = case[-8, ], method = "lm", formula = y ~ x,
              se = FALSE, colour = "#4E79A7") +
  geom_point(aes(colour = row == 8), size = 3) +
  geom_text(aes(label = row), vjust = -1, size = 3) +
  labs(title = "Crash reports vs beta testers online", x = "testers",
       y = "crash_reports")
```

OUTPUT:



Output 4: crash_reports vs testers (row 8 in red; dashed line = all rows, solid line = without row 8)

Measure	Value
Row that looks wrong	Row 8 (testers = 195, crash_reports = 29), the record typed in by a trainee
r for all 14 rows	0.60 (moderate)
r without row 8 (13 rows)	0.15 (weak)
Effect of the single record	r rises by 0.45

Ans:

The wrong record is row 8: it lies far to the right and far above the other 13 points (195 testers and 29 crash reports, when the rest have 47 to 137 testers and 7 to 20 crash reports). With all rows r = 0.60, but without row 8 it falls to 0.15, so this single record created a moderate link out of almost nothing.

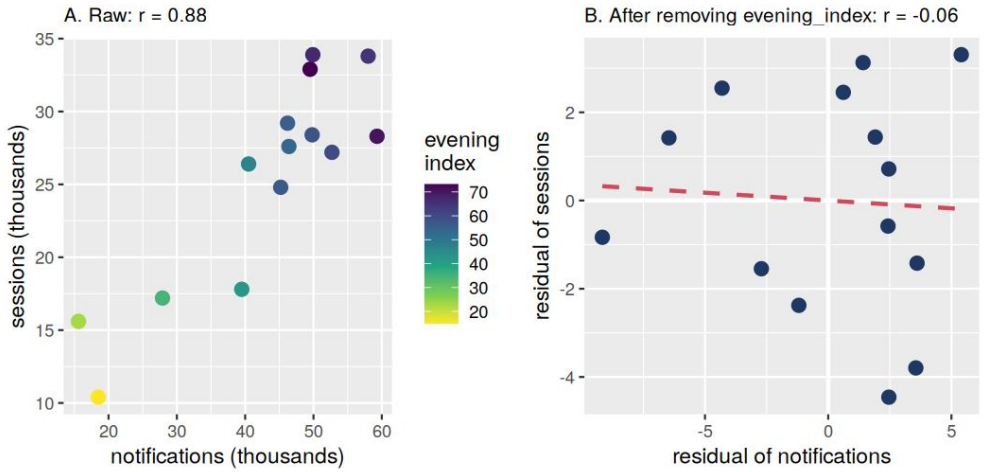
TASK 5: The prime suspect

notifications and sessions are strongly linked. Report r(notifications, sessions), then r(evening_index, notifications) and r(evening_index, sessions). Compute the partial correlation of notifications and sessions controlling for evening_index: correlate resid(lm(notifications ~ evening_index)) with resid(lm(sessions ~ evening_index)). Write a two-sentence verdict: does notifications cause sessions? Name one study design that could test causation.

```
cor(case$notifications, case$sessions)
cor(case$evening_index, case$notifications)
cor(case$evening_index, case$sessions)

res_n <- resid(lm(notifications ~ evening_index, data = case))
res_s <- resid(lm(sessions      ~ evening_index, data = case))
cor(res_n, res_s)                                  # partial correlation
```

OUTPUT:



Output 5: (A) notifications vs sessions coloured by evening_index; (B) the residuals after evening_index is removed

Measure	Value
r(notifications, sessions)	0.88
r(evening_index, notifications)	0.95
r(evening_index, sessions)	0.94
Partial r (notifications, sessions | evening_index)	−0.06

Ans:

Verdict: No, notifications does not cause sessions. The strong link (0.88) disappears once evening_index is removed from both variables (partial r = −0.06), so evening usage is a hidden driver that raises both notifications and sessions. A study that could test causation is a randomised A/B experiment: randomly give some beta testers normal notifications and others reduced notifications for two weeks and compare their sessions, because random assignment balances evening usage between the two groups.

Summary

Task	Trap in the case file	Finding
1	Matrix	n = 14, 21 distinct pairs; one strong block (evening_index, notifications, sessions)
2	Strongest clues	Positive: evening_index and notifications, r = 0.95 (strong). Negative: sessions and testers, r = −0.37 (moderate)
3	A curve that fools r	bright_dev and eye_strain: r = 0.03 but quadratic R-squared = 0.99
4	A single wrong record	Row 8: r = 0.60 with it, 0.15 without it
5	A hidden driver	notifications and sessions: r = 0.88, partial r = −0.06 after controlling for evening_index

Conclusion

The three traps in the case file each break one part of the detective's rule. A zero correlation (bright_dev and eye_strain) does not mean no relationship, because r sees only straight lines. One typed-in record (row 8) was enough to turn a weak link (0.15) into a moderate one (0.60). A strong link (notifications and sessions, 0.88) came from a hidden driver, evening_index, and vanished once it was controlled for (−0.06).

So r should always be read together with a plot, checked for single records, and never taken as proof of cause.